LASER

Light Amplification By Stimulated Emission of Radiation

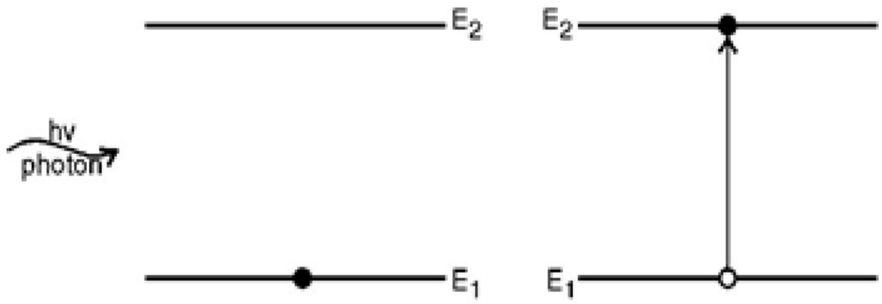
LASER

The word LASER is the acronym for Light Emission by Stimulated Emission of Radiation. As the name suggests in LASER, the light is amplified the help of a process called Stimulated Emission. Thus LASER is based on the principle of Stimulated Emission.

In other words, it is a device to produce a monochromatic beam of light. The theoretical basis for the development of LASER was provided by the great scientist Albert Einstein in 1971 when he emitted the possibility of stimulated emission of radiation.

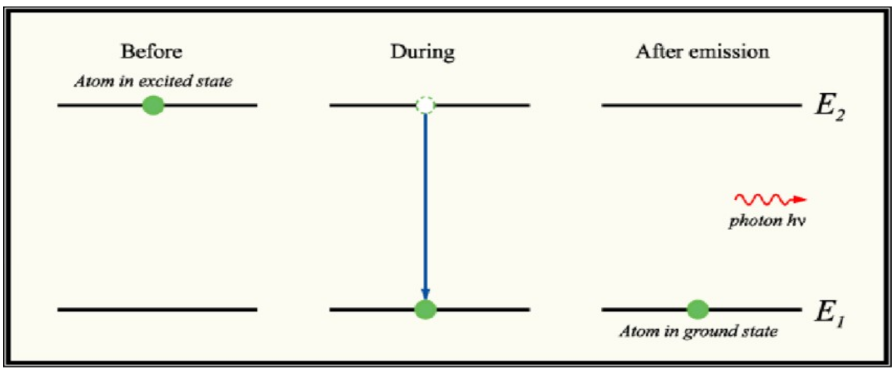
1.0 Stimulated Absorption:

When a photon of light having energy E_2- E_1= hv is incident on an atom in ground state. The atom in the ground state may absorb the photon and jump to higher energy state E_2. This process is called stimulated absorption or induced absorption. This is called so because of incident photon has stimulated the atom to absorb energy.



2.0 Spontaneous Emission:

If the atom in an excited state automatically decays to ground state by emitting photon of energy (E_1-E_2=hv). Then this process is called Spontaneous emission. Generally atom in excited state can stay for 10^{-9} to 10^{-8} seconds.



Characteristics of Spontaneous Emission:

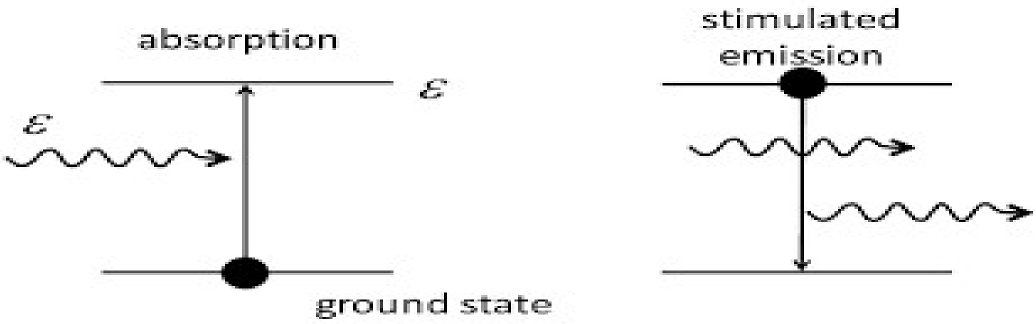
i. The emitted photon of energy (E_1-E_2=hv) can move in any random direction.

ii. There will not be any phase relationship between the photon emitted from various atoms.

Hence, the radiation coming out due to spontaneous emission is incoherent.

3.0 Stimulated Emission:

If the atom in the excited state E_2 and a photon of energy exactly equal to $E_1 – E_2 = hv$ is incident on it, then the incident photon interact with the atom in the excited state then it stimulate and induces the atom to come down to ground state E_1. A fresh photon is emitted in this process. Therefore, when an atom ejects photon due to its interaction with a photon incident on it, this process is called stimulated emission (or induced emission).



Characteristics of Stimulated Emission:

a) For each incident photon, there are two ongoing photons moving in same direction.

b) As the emitted photon has exactly the same energy, phase and direction as incident photon, we will achieve an amplified and unidirectional coherent beam.

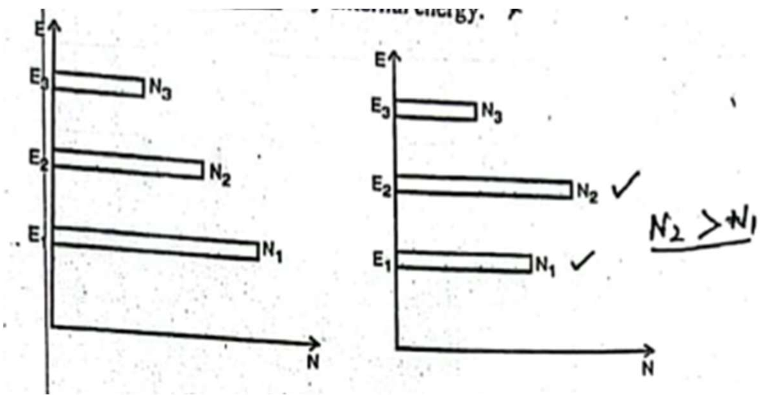
4.0 Population Inversion in LASER

Normally the population density of atoms/electron is more in the ground state than excited state. But if the process of stimulated emission dominates over the process of spontaneous emission, then it may be possible that $N_2 > N_1$. Where N_1 is the number of atom in the ground state and N_2 is the number of atom in the excited state.

This process of achieving greater population density of atoms in higher energy state as compared to lower energy state is called population inversion. The atom in the lower energy state are raised to excited state by external energy.

5.0 Metastable State:

It is also the excited state but having life time 10^{-8} to 10^{-9} seconds. Population inversion occur only in between metastable state and lower state. The two states in between the population inversion occur and LASER is



achieved, the upper one is known as upper LASER level (ULL) and lower on is known as lower LASER level (LLL).

6.0 Components of LASER:

Three main components of LASER devices re:

a) Active medium

b) Pumping source

c) Optical resonator system

6.1 Active Medium of LASER

When the energy is given to LASER medium (solid, liquid, gas), then only a small fraction of LASER shows lasing action. This part of LASER medium is called active medium or active center. Thus due to this reason LASER medium is called heart of LASER. For example in case of ruby LASER Al_2O_3 is doped with Cr_2O_3. This LASER is due to doped chromium ion. Thus Cr^{3+} are active centers.

6.2 Pumping Source

As we discussed previously, the principle of LASER is stimulated emission and for it to take place, Population inversion has to be achieved and maintained. For this there must be a source of external energy, which can continuously supply energy to energy-to-LASER medium, so that population inversion can be achieved. Such a source of energy is called pump or pumping source and the process of applying energy to LASER medium so as to achieved the population inversion is called pumping.

Types of Pumping Source

Depending upon the type of LASER, the most commonly used pumping source are listed below;

a) Optical Pumping:

In this population, inversion is achieved by several means of light energy delivered from appropriate pumping source, such as gaseous discharge or flash tubes. For example in ruby LASER, Xenon flash tube is used.

b) Electrical Discharge:

This type of pumping accomplished by mean of intense electrical discharge in the medium and is particularly suited to gas media like He-Ne LASER and CO_2 LASER. The electric discharge convert the gas into plasma where active centers collides in-elastically with free electrons and population inversion is achieved.

c) Chemical Pumping:

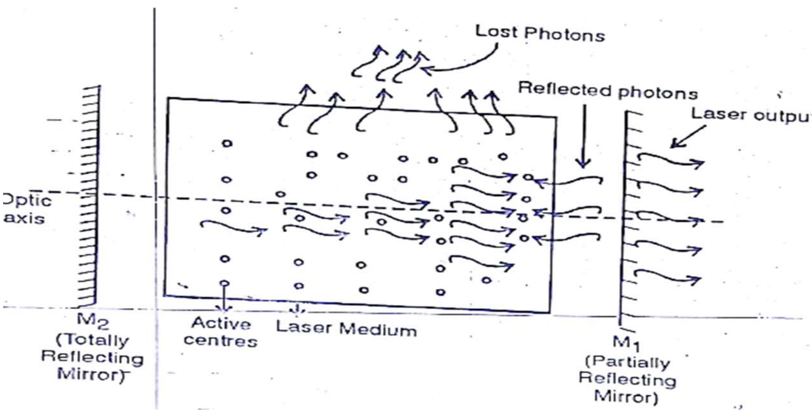
It raises active centers into higher level by mean of suitable exothermal chemical reaction in active medium.

d) Heat Pumping:

In this type of pumping, the active material is first brought to a higher temperature then rapidly cool down.

6.3 Optical Resonator System

An optical resonator is a system or setup which is use to obtain amplification of stimulated photon by oscillating them back and forth between the system of two mirrors. Thus it consist of two plane or concave mirrors. One of the mirror is partially reflecting (having reflectivity less than 100%) and other is totally reflecting (having reflectivity 100%). Laser output is received from partially reflecting mirrors. The space between two mirrors is called cavity.



7.0 Types Of LASER:

The LASER can be classified as various ways, as explained below:

a) **State of LASER medium:** According to state of LASER medium, we have solid state LASER like ruby LASER, gas LASER like He-Ne LASER.

b) **Mechanism of Pumping:** According to the pumping mechanism, we have flash light or optical pumping based LASER like ruby LASER, electric discharge based LASER like He-Ne LASER.

c) **Nature of Output:** According to nature of output, we have pulsed LASER like ruby LASER and continuous wave LASER like He-Ne LASER.

d) **Wavelength of Output:** According to wavelength of LASER, we have ultraviolet, visible and infrared LASER.

8.0 Properties of LASER

A LASER beam has following important characteristics;

- Divergence and directionality: LASER is highly directional beam
- Monochromaticity: LASER is monochromatic that is it has single wavelength
- Brightness, Coherence

8.1 LASER Coherence

When a LASER beam is emitted from a source, there may be the phase difference between the beam of different points that is it may be possible that there may be some time gap between two atoms to come from upper energy level to lower energy level, the term coherence refer to the degree of co-relation between the phases.

In other words, two or more light waves are said if they bear a constant phase relation among themselves.

Coherence can be classified into two ways;

a) **Temporal Coherence:** Consider a light wave travelling along +X axis. Consider two different points A and B along the same wave train that is along +X axis.

If Φ(A) is phase of point A at any time and Φ(B) is phase of point B at any time then phase difference between these points is given by

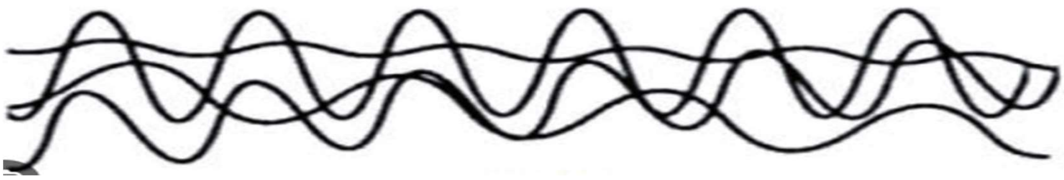
$$\Phi = \Phi(A) - \Phi(B)$$

If fi is independent of time then point A and B are said to exhibit temporal if the phase difference of wave crossing the two points lying on the plane parallel to direction of propagation of beam is independent of time.

b) **Spatial Coherence:** Consider a light wave travelling along +X axis. Draw a line perpendicular to direction of beam. Consider two different points C and D on this line.

If Φ(C) is phase of point C at any time and Φ(D) is phase of point D at any time then phase difference between these point is given by

$$\Phi = \Phi(C) - \Phi(D)$$



If it is dependent of time then point C and D are said to exhibit spatial coherence or transverse coherent or lateral coherence.

In other words, a beam of LASER is said to exhibit spatial coherence if the phase difference of the wave crossing the two points lying on the plane perpendicular to the direction of propagation of the beam is independent of time.

9.0 3-Level LASER

In a three-level LASER, the material is first excited to a short-lived high-energy state that spontaneously drops to a somewhat lower-energy state with an unusually long lifetime, called a metastable state.

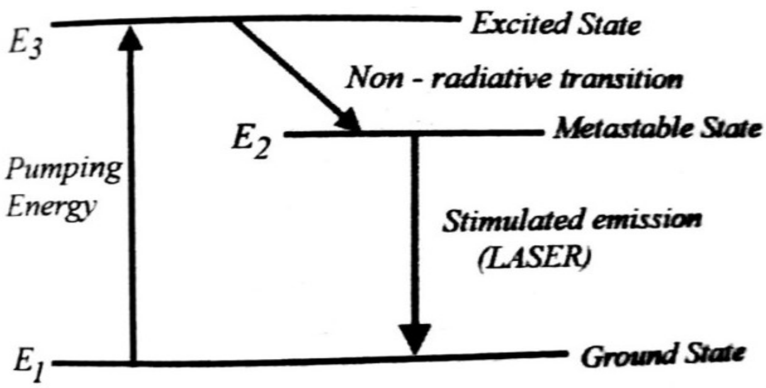
Example:

Ruby LASER is well known example of 3-level LASER.

Principle:

Atoms are pumped from the ground state **E_1** to higher state **E_3** with the help of pumping source. The E_3 is called pumping state. The life time of atoms is least in the energy level E_3. It means E_3 is unstable stable state and here atoms stay for 10^{-9} to 10^{-8} seconds.

9.1 Working:



Consider a system consisting of three energy levels E_1, E_2, E_3 with N number of electrons.

Energy Levels In 3-Level LASER:

We assume that the energy level of E_1 is less than than E_2 and E_3, the energy level of E_2 is greater than E_1 and less than E_3, and the energy level of E_3 is greater than E_1 and E_2.

It can be simply written as $E_1 < E_2 < E_3$. That means the energy level of E_2 lies in between E_1 and E_3 energy levels.

- **Ground State:**

E_1 is known as the ground state or lower energy state and the energy levels **E_2** and **E_3** are known as excited states.

- **Metastable State:**

The energy level **E_2** is sometimes referred to as **Meta stable state**.

- **Pump State:**

The energy level **E_3** is sometimes referred to as **pump state** or **pump level**.

- **Number of Electrons:**

The N number of electrons in the system occupies these three energy levels. Let **N_1** be the number of electrons in the energy state **E_1**, **N_2** be the number of electrons in the energy state **E_2** and **N_3** be the number of electrons in the energy state **E_3**.

Condition for Population Inversion

To get LASER emission or population inversion, the population of higher energy state (E_2) should be greater than the population of the lower energy state (E_1). $\mathbf{N_2}$ **greater than** $\mathbf{N_1}$.

- Under normal conditions, the higher an energy level is, the lesser it is populated. For example, in a three level energy system, the lower energy state E_1 is highly populated as compared to the excited energy states E_2 and E_3.On the other hand, the excited energy state E_2 is highly populated as compared to the excited energy state E_3.

It can be simply written as

$$\mathbf{N_1 > N_2 > N_3.}$$

- Under certain conditions, the greater population of higher energy state (E_2) as compared to the lower energy state (E_1) is achieved. Such an arrangement is called population inversion.

Pumping:

Let us assume that initially the majority of electrons will be in the lower energy state or ground state (E_1) and only a small number of electrons will be in excited states (E_2 and E_3). When we supply light energy which is equal to the energy difference of E_3 and E_1, the electrons in the lower energy state (E_1) gains sufficient energy and jumps into the higher energy state (E_3). This process of supplying energy is called pumping.

$$\mathbf{\Delta E=E_3-E_1}$$

Other Methods of Excitation:

- We also use other methods to excite ground state electrons such as electric discharge and chemical reactions.

10.0 4-Level LASER

In a four-level LASER, **the LASER transition finishes in an initially unoccupied state F, having started in a state I, which is not the ground state**. As the state F is initially unoccupied, any population in I constitutes population inversion. Thus LASER action is possible if I is sufficiently metastable.

Example

The **Nd:YAG (Geusic, Marcos, & Van Uitert, 1964) (neodymium YAG)** LASER is an example of a four level LASER. In this LASER, neodymium (III) ions entrained in a yttrium aluminum garnet crystal provide the four energy levels.

Principle:

The LASER transition finishes in an initially unoccupied state F, having started in a state I, which is not the ground state. As the state F is initially unoccupied, any population in I constitutes population inversion. Thus LASER action is possible if I is sufficiently metastable

Working:

Consider a group of electrons with four energy levels $\mathbf{E_1}$**,** $\mathbf{E_2}$, $\mathbf{E_3}$**,** $\mathbf{E_4}$**.**

E_1 is the lowest energy state, E2 is the next higher energy, E_3 is the next higher energy state after E_2, E_4 is the next higher energy state after E_3.

The number of electrons in the lower energy state or ground state is given by N_1, the number of electrons in the energy state E_2 is given by N_2, the number of electrons in the energy state E_3 is given by N_3 and the number of electrons in the energy state E_4 is given by N_4.

Lifetime

We assume that $E_1 < E_2 < E_3 < E_4$. The lifetime of electrons in the energy E_4 state and energy state E_2 is very less. Therefore, electrons in these states will only stay for very short period.

Electronic Transition:

When we supply light energy which is equal to the energy difference of E_4 and E_1, the electrons in the lower energy state E_1 gains sufficient energy and jumps into the higher energy state E_4.

The lifetime of electrons in the energy state E_4 is very small. Therefore, after a short period they fall back into the next lower energy state E_3 by releasing non-radiation energy.

The **lifetime** of electrons in the energy state $\mathbf{E_3}$ is very **large** as compared to E_4 and E_2. As a result, a large number of electrons accumulate in the energy level E_3. After completion of their lifetime, the electrons in the energy state E_3 will fall back into the next lower energy state E_2 by releasing energy in the form of photons.

Like the energy state E_4, the lifetime of electrons in the energy state E_2 is also very small. Therefore, the electrons in the energy state E_2 will quickly fall into the next lower energy state or ground state E_1 by releasing non-radiation energy.

Population Inversion:

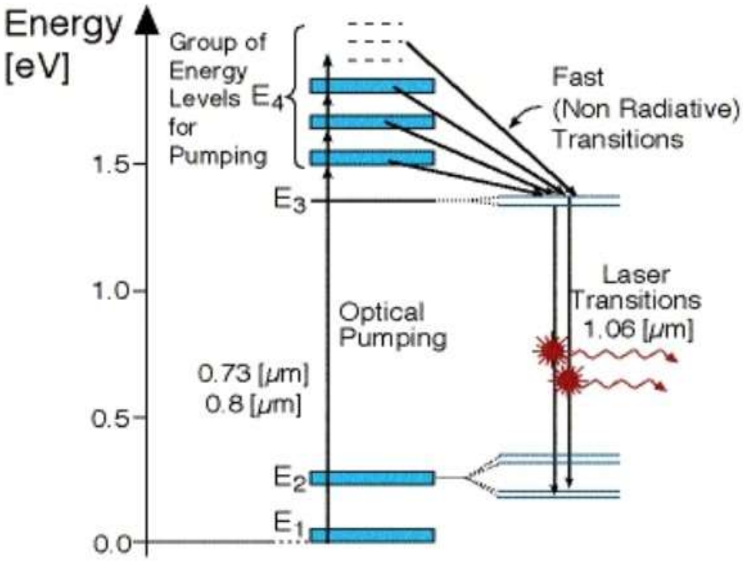
Thus, population inversion is achieved between energy states E_3 and E_2.

Superiority Of 4-level LASER Over 3-level LASER:

In a 4-level LASER, only a few electrons are excited to achieve population inversion. Therefore, a 4-level LASER **produces light efficiently than a 3-level LASER**. In practical, more than four energy levels may be involved in the LASER process.

Frequency Of Incident Photons:

In 3-level and 4-level LASER, the frequency or energy of the pumping photons must be greater than the emitted photons.



Energy Level Diagram

Comparison of 3-Level and 4-Level LASER:

3-Level LASER	4-Level LASER
Requires three energy levels.	Requires four energy levels.
The Output is pulsed.	The output is continuous.
Lasing occurs once.	Lasing occurs at upper and lower levels.
Requires higher pumping rate.	Requires higher pumping rate.
Continuous operation impossible.	Continuous operation is possible.
Ruby LASER is common example.	**He-Ne** and **ND:YAG** are common examples.

ADVANTAGES

3-Llevel LASER	4-Level LASER
Provide an avenue for transitions instead of a truncated.	A 4 level LASER has a lower threshold frequency than a 3 level LASER.
More realistic having high efficiency.	There is also very high efficiency.
Vigorous pumping rate provide suitable output.	A lower pumping rate is needed.
Pulse functioning is characteristic.	Constant functioning is additionally feasible.

DISADVANTAGES

3 Level LASER	4 Level LASER
In 3-level LASER terminal level is the ground level and here more than half of the atoms are transferred to E_2.This requires more pumping power.	In 4-level LASER for energy levels can cause procedure more complicated and lengthy. It also requires higher pumping rate for better output.
Usually Output is not continuous but in pulse form.	Usually Output is continuous but continuous operation requires high energy input.
Population Inversion can't be achieved easily.	Population Inversion can't be achieved easily.
Lasing occurs once.	Lasing occurs twice but input is higher.

11.0 NITROGEN LASER

A nitrogen LASER is a gas LASER operating in the ultraviolet range (typically 337.1 nm) using molecular nitrogen as its gain medium, pumped by an electrical discharge. In nitrogen LASER, active or gain or lasing medium is nitrogen molecules in the gaseous phase. It is a three-level LASER.

Discovery:

In 1960, while working at Bell Laboratories, Javan invented the world's first gas LASER. Nitrogen LASER were first developed in 1963, and began to be used commercially in 1972.

Principle:

Nitrogen LASER works on the base of strong electrical discharge through nitrogen gas. The nitrogen gas can be supplied through a gas cylinder or from liquid nitrogen. The LASER light is emitted in UV range with a short pulse width and high intensity. It uses electricity to excite nitrogen. When an electric spark crosses a spark gap in the LASER, the electrons heat up the nitrogen atoms in air by exciting them into metastable state. When a photon of wavelength 337nm passes the excited nitrogen, a stimulated emission occurs and a LASER state is generated.

Properties of N_2 Laser:

Type: Gas LASER

Pump source: Electric discharge

Operating wavelength: 337.1nm

Gain medium: Gaseous N_2

Range: UV region

Construction and Working of Nitrogen Laser:

It consists of a tube formed by two electrodes connected to the discharge voltage source (~ 10-40 KeV). The active medium in the form of Nitrogen molecules in the gaseous phase is placed between these two electrodes.

The arrangement is enclosed in a resonant cavity formed by two planes and parallel mirrors M_1 and M_2. Mirror M_1 is a perfectly reflecting mirror, while mirror M_2 is a partially reflecting mirror through which pulse the LASER comes out.

The Active Medium

The main component of a Nitrogen LASER is a medium in the form of Nitrogen molecules called an active medi0um. The main characteristics of the active medium are as follows:

- It must have a pair of energy levels separated by a certain amount of energy. The energy level having energy is known as an upper energy level or higher excited energy level and the energy level having low energy is known as low energy or ground state.
- It must allow a population inversion between two energy levels.

Pumping Device

It is an external source of energy that provides the necessary energy to the active medium to produce a state of population inversion essential for lasing action. In nitrogen LASER, active gain or LASER medium is nitrogen molecules in gaseous phase. Population inversion in nitrogen LASER is achieved by an electrical pumping source.

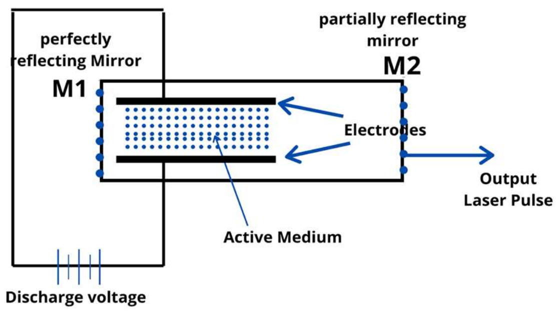
A Resonant Cavity

Population inversion is achieved to amplify the signal by (or photon) stimulated emission. However, in practice, most of the atoms in the excited state emit spontaneously and do not contribute to the overall output .Only a few atoms in the excited state emit via stimulated emission and hands overall gain of the output is small. Therefore, we require a positive feedback mechanism to make most of the atom in the excited state to emit via stimulated emission for contributing to the current output. Thus, a resonant cavity or resonator is a feedback device that makes the photon to move back and forth through the active medium. In this process, the number of photons emitted due to stimulated emission are multiplied.

A resonant cavity consists of a pair of plane or spherical mirrors placed parallel to each other at the end of the active medium. One of the mirrors is a fully reflecting mirror and the other is a partially transmitting mirror .The LASER output is taken out through the partially transmitting mirror which is also called the output coupler mirror. An optical resonator is needed to build up the light energy in the beam. The resonator is formed by placing a pair of mirrors facing each other so that light emitted along the line between the mirrors is reflected back and forth. When a population inversion is created in the medium, light reflected back and forth increases in intensity with each pass through the LASER medium.

The combination of LASER medium and resonant cavity forms what often is called simply a LASER but technically is a LASER oscillator. Oscillation determines many LASER properties, and it means that the device generates light internally.

Population Inversion in Nitrogen LASER:



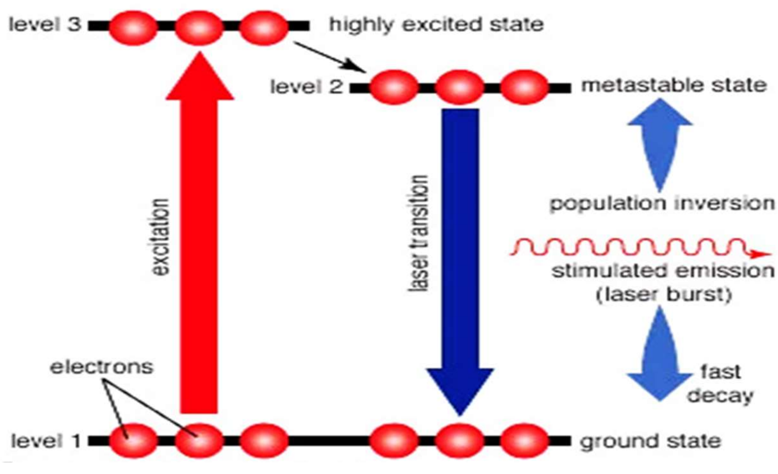
Population inversion in nitrogen LASER is achieved by electric discharge pumping. In this case, voltage is applied across the electrodes of the gas discharge tube which is filled with a low-pressure gas mixture known as the gain medium.

The applied voltage produces an electric field within the tube. This electric field accelerates electrons within the gas. These electron collides with the gas atom or gain medium and excite their atom to higher energy levels or excited energy levels.

If the atom in the lower-lying energy level makes the transition to the excited state faster than the atom in the higher-lying energy level makes the transition to the lower-lying energy levels, then the population of atom in the higher energy level is more than the population of atom in lower energy levels. Hence, population inversion in gases is achieved.

Energy Level Diagram of Nitrogen LASER:

The energy level diagram of the Nitrogen molecule in the gaseous phase is shown in the figure. Each energy level consists of a series of vibrational energy levels depending on the inter-nuclear separation.



Nitrogen molecule is excited from the ground state E_1 to the excited energy level 3 (called upper LASER label ULL) by direct collision with electrons in the discharge tube. Nitrogen molecule falls from energy level E_3 energy level E_2 (called lower

LASER level LLL) by emitting a photon of ultraviolet light of wavelength 337.1 nm. The lifetime of the upper energy level is less than the lifetime of the lower energy level. The output of the Nitrogen LASER is in the form of pulses i.e., nitrogen LASER is a pulsed LASER.

In Nitrogen LASER, the width of the LASER pulse is narrow. It is because when a LASER transition takes place, the population of upper LASER level E_3 decreases, and the population of lower LASER level E_2 increases. The nitrogen molecule then falls from energy level E_2 to the ground state and Laser action stops. That is why nitrogen LASER is known as **self-terminating LASER.**

Advantages:

- Well understood
- Relatively cheap gain medium
- Large volumes of active material
- Very Efficient
- Fragile

Disadvantages

- The Beam quality of the Nitrogen LASER is poor.
- The divergence of the Nitrogen LASER is large as compared to the divergence of other LASER.
- The output power is low.
- The efficiency of a Nitrogen LASER is low.

Applications of a Nitrogen LASER:

- LASER deliver coherent, monochromatic, well-controlled, and precisely directed light beams.
- Nitrogen LASER can be used for a wide range of applications in the UV-visible region.
- They can be easily coupled to a microscope for carrying out experiments in life science laboratories.
- Efficient sources for LASER-induced fluorescence, photochemistry and general spectroscopy.

Also used in:

- Measuring air pollution.
- Scientific research.
- Spectroscopy and fluorescence studies.
- Fast-speed photography

- Treatment of non-healing wounds, pulmonary tuberculosis, etc
- Transverse optical pumping of dye LASER.

12.0 Alexandrite LASER

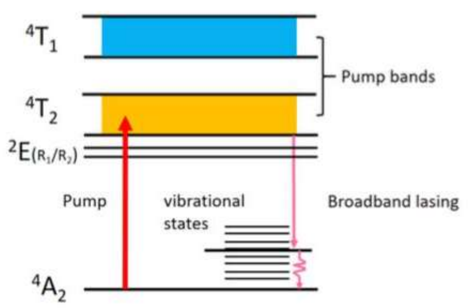
Alexandrite is the common name for Chromium-doped chrysoberyl (Cr^{3+}: $BeAl_2O_4$). Solid-state LASER based on alexandrite (Cr^{3+}: $BeAl_2O_4$) crystals. The observed color of Alexandrite changes from greenish blue under daylight to red under incandescent illumination. Alexandrite is of great interest as a solid-state LASER material since it has a broad emission band (701-858 nm) and a high intrinsic slope efficiency of 65%. It is tunable solid state LASER.

Properties of Alexandrite LASER	
Density	3.7(g/cm^3)
Refractive index	1.74 (750 nm)
Thermal conductivity	0.23 (W/cmK)
Doping concentration	0.05%-0.3%
Melting point	1870 °C
Chemical formula	(Cr3+: BeAl2O4)
Crystal structure	Orthorhombic

Lasing Mechanisms

Alexandrite can be operated as a 3-level system or a 4-level vibronic LASER. Two theoretical models have been well developed for describing the different kinetics. As a 3-level LASER system, the energy level structure of Alexandrite LASER is similar to that of Ruby LASER. The simplified energy level diagram of Chromium ions in Alexandrite is shown.

The $^4A_2$ level is the ground state. The vibronically broadened $^4T_1$ and $^4T_2$ levels, arising from the strong coupling between the outermost unshielded electrons of the transition ions Cr^{3+}, are two adjacent absorption state energy continua. Similar to Ruby, Alexandrite has a metastable level ^2E, which acts as a storage level. Ions excited to the $^4T_1$ and $^4T_2$ states rapidly decay to the ^2E state, and the lifetime of the ^2E state (~1.5 ms) is much longer than that of the $^4T_1$ and $^4T_2$ (~6.6 µs) states. The ^2E level is actually a doublet that consists of R_1 (680.4 nm) and R_2 (678.5 nm) lines. The emission cross section at 680.4 nm is significantly larger than that at 678.5 nm. Therefore, lasing at the vicinity of the 680.4 nm wavelength can be obtained utilizing the transition from the ^2E level to the $^4A_2$ ground level, which corresponds to the R-line emission lasing at 694.3 nm of the Ruby LASER.

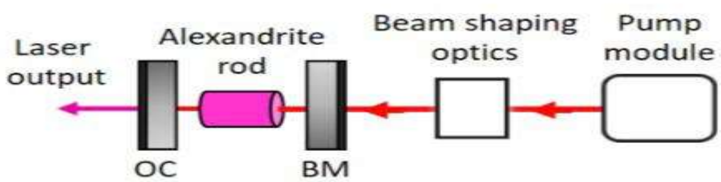


In the 3-level mode, the metastable ^2E state is regarded as the upper LASER level, whereas the $^4A_2$ level is viewed as the lower LASER level. As a **3-level system**, Alexandrite has a high LASER threshold, low efficiency and fixed output wavelength of 680.4 nm at room temperature.

Alexandrite is of interest primarily because of its vibronic nature. For the 4-level mode, transitions from the upper LASER level $^4T_2$ to vibrationally excited states within $^4A_2$ results in vibronic lasing. The vibrational states are higher than the ground level. The upper LASER level $^4T_2$ is always in thermal equilibrium with the ^2E level since the energy difference (800 cm^{-1}) between $^4T_2$ and ^2E. A considerable population will be present in the $^4T_2$ state when the ^2E state has been populated. By contrast, the energy difference between the $^4T_2$ and ^2E levels in Ruby is 2300 cm^{-}. Hence, effectively all the excited populations are stored in the ^2E level. The LASER wavelength of Alexandrite depends on which vibrational state acts as the terminal level of the transition. Therefore, the transition occurring between two broad energy bands results in broadly tunable radiation of Alexandrite LASER. Additionally, the fact that the ^2E and $^4T_2$ levels are in thermal equilibrium in Alexandrite is a critical reason why the performance of Alexandrite LASER is temperature-dependent.

Components of Alexandrite Laser

A compact cavity was developed and investigated to assess the power and efficiency potential of the diode-end-pumped Alexandrite LASER. The LASER system consisted of a pump module, a pump beam shaping system, a compact cavity and an Alexandrite rod. The pump module used was the high-power free-space diode LASER.



1.0 LASER-DIODE PUMP SOURCE

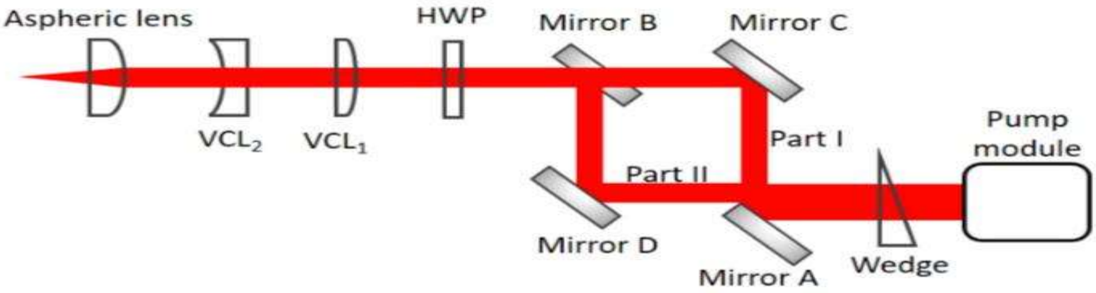
High power red diode LASER have become commercially available in recent years. Two commercial LASER diode modules were used as pump sources in this work:

- a tailored free-space LASER module
- a fiber-coupled one

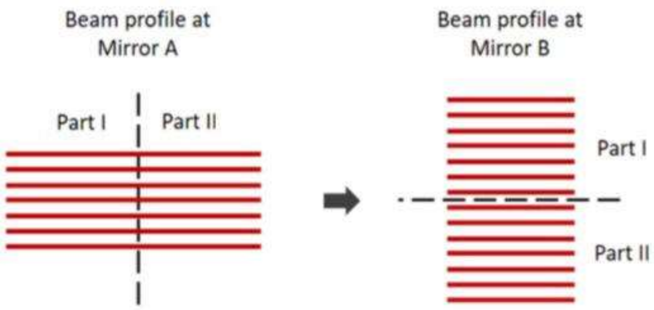
The free-space LASER-diode module provides high output power, high polarization purity, but low beam quality as compared with the fiber-coupled one.

2. PUMP BEAM SHAPING

The pump beam was spatially redistributed with a beam shaping system to improve the horizontal beam quality but at the expense of worsening the vertical one. The highly multimode spatial output of the diode module was reshaped by the optical system.



The pump beam firstly hits a wedge that deflects the beam at a small downward angle (4°). The concept of using four mirrors to split and re-stack the pump beam through the wedge. Two edge coated mirrors (Mirror A and B) are highly reflecting at the pump wavelength (~638 nm) at 45°, and the other two mirrors (Mirror C and D) are HR at visible wavelengths at 45°. Mirror A horizontally splits the beam into two parts - Part I and II. At the point of splitting, the height of Part I is the same as that of Part II. Part II continues downwards until it reaches Mirror D, whereas Part I that reflected by Mirror A remains the same height. Part I and Part II, respectively reflected by Mirror C and D, were re-combined one above the other in the vertical direction at Mirror B.



Schematic diagram of splitting and re-stacking the pump beam

The re-stacked beam then passed through a Galilean telescope system formed of two vertical cylindrical lenses (VCL1 & VCL2) with focal lengths of 150 and -50mm.This system was used to narrow the pump beam in the vertical direction. A half-wave plate (HWP) was used to alter the polarization state of the pump beam to be parallel with the b-axis of the Alexandrite rod for high absorption. The pump beam was finally focused by an aspherical lens with 40 mm focal length to a spot.

3. **Alexandrite Rod**

The end-pumped Alexandrite rod with a length of 10 mm, a diameter of 4 mm and 0.22 at. % Cr^{3+} concentration was embedded in a water-cooled copper heat-sink for temperature control. The end faces of the Alexandrite rod were plane-parallel and anti-reflection coated at LASER wavelength (~755 nm). The absorption coefficient of the alexandrite rod was measured to be 6.0 cm^{-1} with a He-Ne LASER (633 nm) for light polarized parallel to the b-axis of the crystal. Approximately 5% of the incident pump was reflected by the coated plane surface of the Alexandrite rod, and the remaining 95% was almost completely absorbed by the crystal.

4. **Compact Cavity**

The compact cavity was formed of two plane mirrors: a dichroic back mirror (BM) which was highly-transmitting (R99.9%) at LASER wavelength (~755 nm) and an output coupler (OC). The compact cavity was investigated with three output couplers with reflectance of 99%, 98% and 97%, respectively. The cavity length was kept at 16mm.

Applications

- **i.** Alexandrite LASER are used for treating skin disorders (e.g., pigmented lesions).
- **ii.** For hair removal, its clinical use is well established because of its suitable wavelength, which is in the midrange of the melanin-absorbing spectrum and targets the melanin of the hair.
- **iii.** They are also effective in the treatment of unwanted hair, as the LASER energy is absorbed specifically by the hair follicle rather than any other target on or within the skin surface.
- **iv.** Tattoo removal treatment is used to remove black, blue and green pigment since these colors can effectively absorb the wavelength emitted by Alexandrite LASER.
- **v.** In medicine, alexandrite LASER have been used especially for dermatological procedures, nevus of Ota removal and leg vein treatment.
- **vi.** It is also used as calculus in dentistry.
- **vii.** it is also beneficial in the removal of vascular lesions.

13.0 DYE LASER

Dye LASER was first discovered by **Sorokin** and his **colleagues.** Dye LASER is a liquid LASER in which we use organic dye e.g: Polymethine dye etc. Dye LASER are used in place of solid state LASER because solid state materials has some defects and imperfections.

Compared to gases and most solid state lasing media, a dye can usually be used for a much wider range of wavelengths.

Laser generation has developed many hundred dye LASER but most widely used dye is **Rhodamine 6G (Xanthene)** which emits in yellow – red region.

Dye LASER wavelength Range:

Different dyes have different emission spectra or colors thus allowing dye LASER to cover a broad wavelength range from the ultraviolet (320 nm) to the infrared at about 1500 nm. A unique property of dye LASER is the broad emission spectrum (typically 30–60 nm) over which gain occurs. Some of the Organic dyes are

Rhodamine 6G (orange, 540–680 nm) also known as Xanthene dye

Fluorescein (green, 530– 560 nm)

Coumarin (blue 490– 620 nm)

Stilbene (violet 410– 480 nm)

Umbelliferone (blue, 450– 470 nm), Tetracene, malachite green, and others

Construction:

The dye LASER consisted of a 1 cm long quartz glass tube filled with solutions of organic dyes such as Rhodamine 6G .

Usually LASER consists of three main components.

- Active medium
- Pumping source
- Optical resonator system

Active Medium:

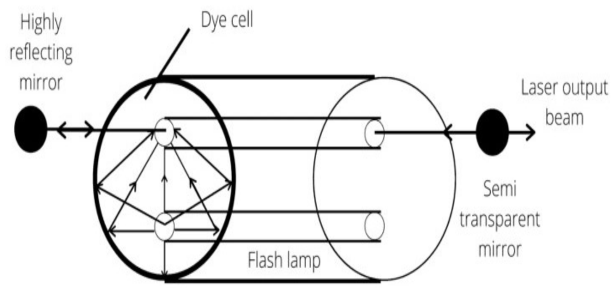
The active LASER medium (also called gain medium or lasing medium) is the source of optical gain within a LASER. The gain results from the stimulated emission of photons through electronic or molecular transitions to a lower energy state from a higher energy state previously populated by a pump source.

In dye LASER, Rhodamine 6G dissolved in a suitable liquid like water, ethyl alcohol or methanol is used as the active medium. The dye solution used in dye LASER typically has a concentration in the range of 10^{-2} to 10^{-4} M. Rhodamine -6G emits in yellow red region.

Pumping Source:

The process of raising large number of atoms from lower energy to a higher energy level is called pumping. So basically, Pumping is the process to achieve population.

There are different types of pumping source used to gain population inversion:



- **Chemical pumping**
- **Thermal pumping**
- **Optical pumping**

In dye LASER, optical pumping source is used. Energy to excite the dye is supplied by a strong light source that may be an arc lamp or another LASER like N_2 LASER or Argon- ion LASER.

Thus, optical pumping is used to excite the dye and to achieve population inversion.

Optical Resonator System:

The most useful feature of dye LASER is their **tunability.** The tunability means that the lasing wavelength for a dye may be varied over a wide range. Due to this reason, dye LASER are also called **tunable LASER.** Tuning over 500 Angstrom has been obtained.

One of the mirror of usual LASER is replaced by **diffraction grating** in dye LASER. A source light is focused onto the dye to excite it and stimulate LASER action.

By rotating the diffraction grating, wavelength of LASER output can be altered. Thus tuning is obtained. Therefore, this combination of partially reflective mirror and diffraction grating will act as optical resonator system. For radiation to be reflected back along the LASER cavity axis, the angle θ that the normal to the diffraction grating makes with the cavity must satisfy the condition.

$$2d\sin\theta = n\lambda \quad (n = 1, 2, 3, \ldots)$$

By rotating the grating, angle θ will be changed and thus the output wavelength will be changed that is tuning of output wavelength will be achieved.

Working:

To operate a dye LASER, an elliptical resonant cavity or resonator is used. The dye cell is placed at one of the foci and the flash lamp is placed at the other foci in the elliptical resonant cavity. The light emitted by the flash lamp is focused on the cell.

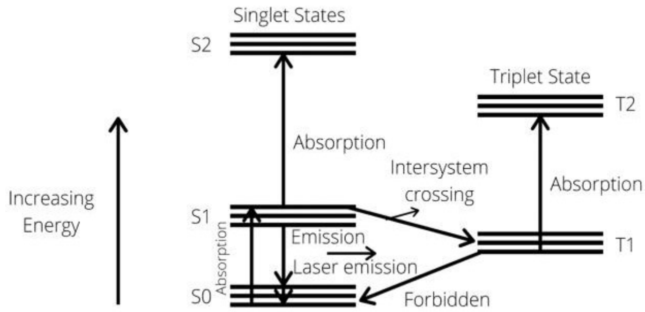
The lights falling on the dye cell causes stimulated emission inside the dye. The emission of radiation due to stimulated emission exists in all directions but the radiation is only amplified along the axis of the cavity formed between highly reflecting mirror and semi-transparent mirror.

Tuning of Dye LASER can be done using various techniques: One of the commonly used techniques is to send a selective wavelength through the Dye. In this case, the totally reflecting mirror of the cavity is replaced by the diffraction grating.

The light from a Nitrogen LASER is made to fall on the front window of the dye cuvette. The concentration of the dye is adjusted in such a way that all the light falling on it is absorbed within a few Millimeters of the front window of the dye cuvette.

The fluorescence from this small area is then reflected back into the dye cell by the diffraction grating and the partially reflecting mirror. As a result, Laser emission takes place. The wavelength of the dye LASER is selected by adjusting the diffraction grating with the help of a micrometer.

This wavelength of light amplifies in the resonant cavity formed by the diffraction grating and partially reflecting mirror. The desired wavelength comes out of the partially reflecting mirror which is refracted by the completely reflected mirror. Hence, the desired wavelength of Dye LASER is obtained.



Advantages of Dye LASER:

- Beam diameter is very less.
- High output power.
- Higher efficiency of 25%.
- Construction is very simple.

Disadvantages of Dye LASER:

- The cost is very high.
- It has complex chemical formula.
- Volatile solvents.
- Short dye life-time.

Applications:

- The most important application of these tunable LASER is the separation of isotopes.
- It has number of applications in the field of holography, spectroscopy and biomedical science.
- It is used to remove the ink of tattoos.
- It is used to remove birth mark of newly born babies.

14.0 HELIUM NEON LASER

Definition:

Helium neon LASER is a type of gas LASER in which a mixture of helium and neon gas is used as a gain medium. Helium Neon LASER is also known as He-Ne LASER.

What is a gas LASER?

A gas LASER is a type of LASER in which a mixture of gas is used as the active medium or LASER medium. Gas LASER are most widely used LASER.

Properties:

- The helium neon LASER is a relatively low power device with an output in the visible red portion of the spectrum.
- The most common wavelength produced by He Ne LASER is 632.8nm.
- For He-Ne LASER the typical LASER tube is from 10 to 100cm in length and the lifetime of such a tube can be as high as 20,000 hours.
- He-Ne LASER are most widely used gas LASER.
- Construction of helium neon LASER is not very complex.

Principle:

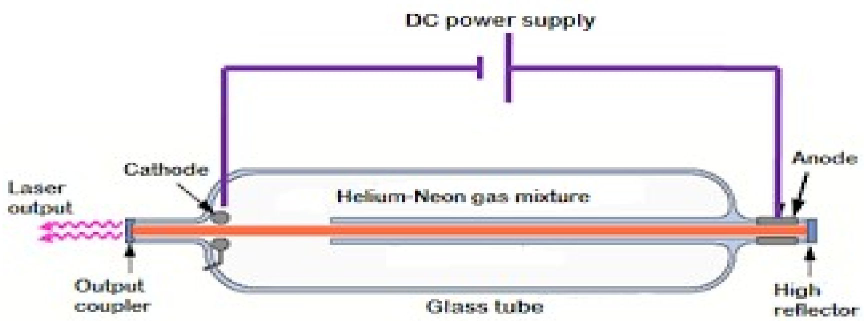
A helium neon LASER is a type of gas LASER whose high energetic medium gain medium consists of a mixture of 10:1 ratio of helium and neon at a total pressure of about 1 torr inside of a small electric discharge. The best known and most widely used helium neon LASER operates at a wavelength of 632.8nm, in the red part of the visible spectrum.

Construction:

The helium neon LASER consists of three essential components:

- Pump source
- Gain medium(LASER glass tube or discharge glass tube)
- Resonating cavity

Pump Source



In order to produce the LASER beam, it is essential to achieve population inversion. Population inversion is the process of achieving more electrons in the high energy state as compared to low energy state.

In general low energy state has more electrons than the high energy state. However, after achieving population inversion more electrons will remain in the high energy state than the low energy state.

In order to achieve population inversion, we need to supply energy to the gain medium or active medium. Different types of energy sources are used to supply energy to the gain medium.

In helium Neon LASER, light energy is not used as pump source. A high voltage DC supplies electric current through the gas mixture of helium and neon.

Gain Medium

The gain medium of a helium neon LASER is made up of the mixture of helium and neon gas contained in glass tube at a low pressure. The partial pressure of helium is 1 mbar whereas that of neon is 0.1 mbar.

The gas mixture is mostly comprised of helium gas. Therefore, in order to achieve population inversion, we need to excite primarily the low energy state electrons of the helium atoms.

In He Ne LASER, neon atoms are active centers and have high energy levels suitable for LASER transitions while helium atoms helium atoms help in exciting neon atoms.

Electrodes (anode and cathode) are provided in glass tube to send the electric current through the gas mixture. These electrodes are connected to a DC power supply.

Resonating Cavity:

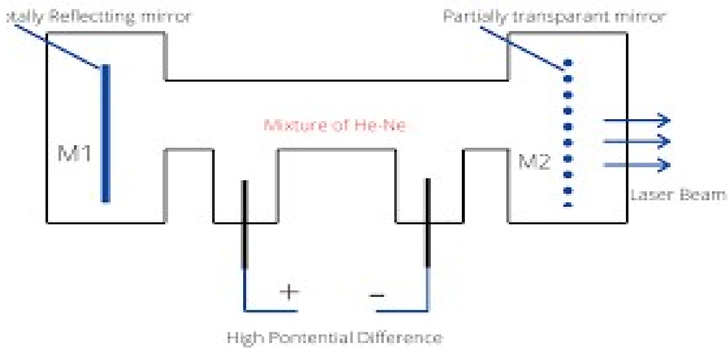
The glass tube containing a mixture of helium and neon gas is placed between two parallel mirrors. These two mirrors are silvered and optically coated.

Each mirror is silvered differently. The left side mirror is partially silvered and is known as output coupler whereas right side mirror is fully silvered and is known as the high reflector or fully reflecting mirror.

The fully silvered mirror will completely reflect the light whereas the partially silvered mirror will reflect the most part of the light but allows some part of the light to produce LASER beam.

Working:

In order to achieve population inversion, we need to supply energy to gain medium. In helium neon LASER, we use high voltage DC as the pump source. A high voltage DC produces energetic electrons that travel through the gas mixture.



The gas mixture in helium neon LASER is mostly comprised of helium atoms. Therefore, helium atoms observe most of the energy supplied by high voltage DC.

When the power is switched on, a high voltage is supplied across the gas mixture. This power is enough to excite the electrons in the gas mixture. The electrons produced in the process of discharge are accelerated between the electrodes through the gas mixture.

In the process of flowing through the gas, the energetic electrons transfer some of their energy to the helium atoms in the gas. As a result, the lower energy state electrons of the helium atoms gain enough energy and jumps into the excited states or metastable states are, the metastable state electrons of the helium atoms cannot return to ground state by spontaneous emission. However, they can return to ground state by transferring their energy to the lower energy state electrons of the neon atoms.

The energy levels of some of the excited states of the neon atoms are identical to the energy levels of the metastable states of the helium atoms.

Unlike the solid, a gas can move or flow between the electrodes. Hence, when excited electrons of helium atoms collide with the low energy state electrons of the neon atoms, they transfer their energy to the neon atoms. As a result, the lower energy state electrons of the neon atoms gain enough energy from the helium atoms and jumps into the higher energy state or metastable states whereas the excited electrons of the helium atoms will fall into the ground state. Thus, helium atoms help neon atoms in achieving population inversion.

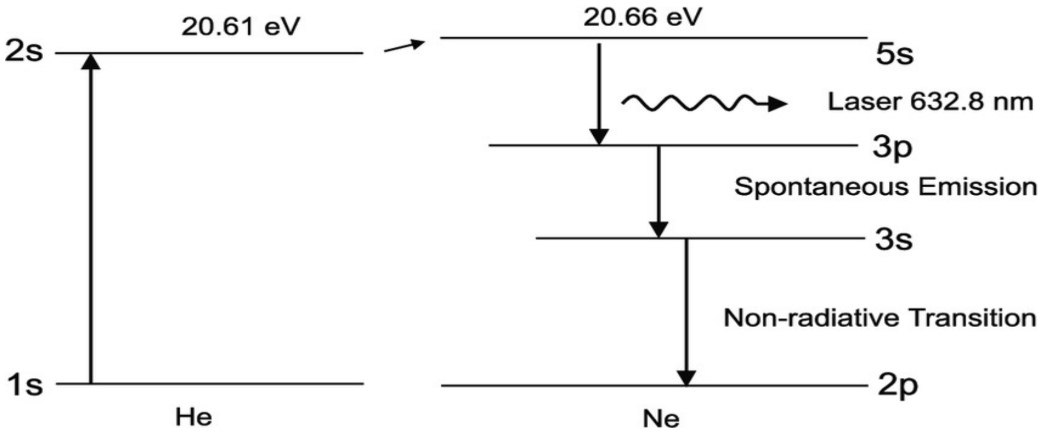
Likewise, millions of ground state electrons of neon atoms are excited to the metastable states. The metastable states have longer lifetime. Therefore, a large number of electrons will remain in the metastable states and hence population inversion is achieved.

After some period, metastable electrons of the neon atoms will spontaneously fall into the next lower energy state by releasing photons or red light. This is called spontaneous emission.

The neon excited electrons continue on to the ground state through radiative and non-radiative transitions.

The light or photons emitted from the neon atoms will moves back and forth between two mirrors until it stimulates other excited electrons of neon atoms and causes them to emit light. Thus optical gain is achieved. This process of photon emission is called stimulated emission of radiation.

The light or photons emitted due to stimulated emission will escape through the partially reflected mirror to produce LASER light.



Advantages:

- Helium neon LASER emits LASER light in the visible portion of the spectrum.
- High stability
- Low cost
- Operates without damage at higher temperature.
- Highly monochromatic

Disadvantages:

- Low efficiency
- Low gain
- He Ne LASER are limited to low power tasks.
- As internal mirrors are used in He Ne LASER to act as optical resonators, but these mirrors are usually eroded by the gas discharge and have to be replaced.

Applications:

- Helium neon LASER are used in industries.
- Helium neon LASER are used in scientific instruments.
- Helium neon LASER are used in the college laboratories.
- Blood analysis.
- Used in tool alignments.
- Are used in reading bar codes which are imprinted on products in stores.

- In LASER printing He Ne LASER is used as a source for writing on the photosensitive material.
- Mainly used in making holograms.

15.0 RUBY LASER

Ruby LASER is a solid-state LASER that was developed by Maiman in 1960 using Ruby as an active medium. Ruby is a crystal of aluminum oxide. In which a part of the aluminium ion is substituted by chromium ion.

Principle:

A ruby LASER works in the principle of **population inversion**. It is the phenomenon done by optical pumping in which the atoms in the ground state are excited to the higher energy state that is metastable state by optical pumping .To achieve population inversion two conditions must be fulfilled:

1-There must be higher number of electrons in metastable state than ground state.

2-Electrons must excited to metastable state.

Parts of LASER:

A ruby LASER consists of three main parts:

- An active material (or LASER medium).
- Optical resonator
- Xenon flash lamp(Pumping source)

Construction:

In ruby LASER, ruby rod of 4cm long and 5mm in diameter is used.

Active Medium:

The active material in the Ruby is chromium ions, obtained by doping a small amount of Chromium oxide (0.05% by weight) in Aluminum oxide so that the chromium ion is replaced by Al ions. These Chromium ions acts as active medium.

Optical Resonator:

Both ends of the rod is highly polished and made strictly parallel. One end is fully reflected and the other one is half polished. These two silvered coated ends of the rod made a optical resonator cavity.

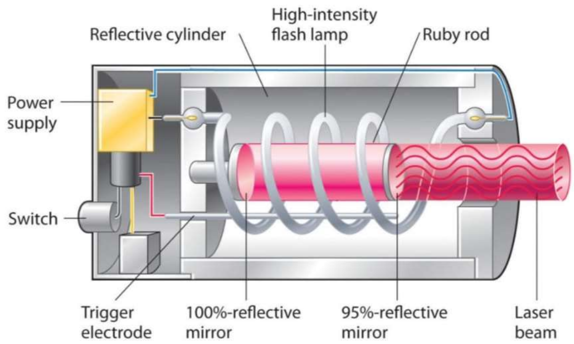
Pumping Source:

The ruby rod is surrounded by xenon flash lamp. This flash lamp provide optical pumping to excite the chromium ions to the upper energy level. Population inversion is achieved by pumping source.

Cooling system:

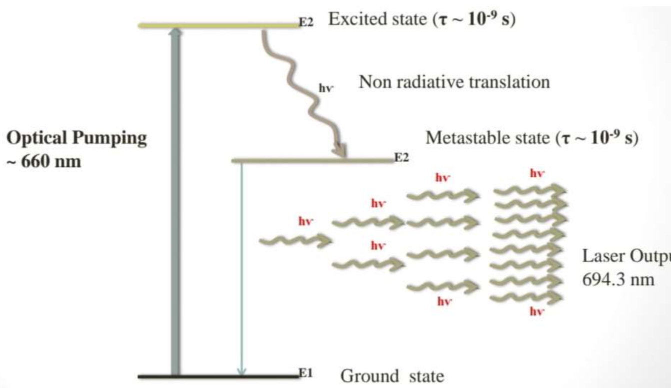
Cooling must be provided for the maintenance of temperature and to protect the instrument from damage caused by overheating.

Working:



Ruby is a three-level LASER system. To understand the working of ruby LASER we have to understand the energy level diagram of chromium ion. Suppose there are three levels E_1, E_2, and (E_3 & E_4). E1 is the ground level, E_2 is the metastable level and E_3 and E_4 are the bands. E_3 & E_4 are considered as only one level because they are much closed to each other.

Pumping:



The ruby crystal is placed inside a xenon flash lamp and the flash lamp is connected to a capacitor which discharges a few thousand joules of energy in a few milliseconds. A part of this energy is absorbed by chromium ions in the ground state. Thus optical pumping raises the chromium ions to energy levels inside the bands E_3 and E_4. This process is called stimulated absorption. The transition to bands E_3 and E_4 are caused by absorption of radiations corresponding to wavelengths approximately 5500 nm.

Achievement of Population Inversion:

Cr^{3+} ions in the excited state lose a part of their energy during interaction with crystal lattice and decay to the metastable state E_2. Thus, the transition from excited states to metastable states is a non-radiative transition or in other words, there is no emission of photons. As E_2 is a metastable state, chromium ions will stay there for a longer time. Hence, the number of chromium ions goes on increasing in the E_2 state, while due to pumping, the number in the ground state E_1 goes on decreasing. As a result, the number of chromium ions becomes more in an excited state (metastable state) as compared to ground state E_1. Hence, the population inversion is achieved between states E_2 and E_1.

Achievement of LASER:

Few of the chromium ions will come back from E_2 to E_1 the process of spontaneous emission by emitting photons. The wavelength of a photon is 6943 Å. This photon travels through the ruby rod and if it is moving in a direction parallel to the axis of the crystal, then it is reflected to and fro by the silvered ends of the ruby rod until it stimulates the other excited ions and causes it to emit a fresh photon in phase with the stimulating photon. Thus, the reflections will result in stimulated emission and it will result in the amplification of the stimulated emitting photons. This stimulated emission is the LASER transition.

The two stimulated emitted photons will knock out more photons by stimulating the chromium ions and their total number will be four and so on. This process is repeated again and again, thus photons multiply. When the photon beam becomes sufficiently intense, then a very powerful and narrow beam of red light of wavelength 6943 Å emerges through the partially silvered end of the ruby crystal.

Output:

The output wavelength of the ruby LASER is 694.3 nm and output power is 10^6 watts and it is in the form of pulses. It operates in pulse mode and hence called pulsed LASER. It is all about the Ruby LASER construction and working.

Advantages:

- It has large power output.
- The pumping efficiencies can be increased by using cylindrical Mirrors.
- It has a narrow linewidth.
- Chemically stable and good thermal conductivity.

Disadvantages:

- The output LASER beam is not continuous but the light is emitted in pulses.
- The Monochromaticity may be affected by crystalline Imperfection. Thermal distortion and scattering.
- Frequent cooling is required as a lot of heat energy is produced during its operation.
- A large amount of energy is required to Trigger LASER oscillations.

Applications:

Following are applications of ruby LASER.

❖ **Industrial use**

- To drill tiny holes in hard materials.
- Welding and machining.

- Highly localised heating.

❖ **In everyday life**

- Bar-code reader
- Compact disc player
- To produce holograms

❖ **Surgical uses**

- To break gallstones and kidneys
- To weld broken tissues
- To destroy cancerous cells
- To remove plaque clogging human arteries.

16.0 Use of Laser in Absorption Spectroscopic Method

Laser absorption spectroscopy (LAS) is an absorption spectroscopic method that employs a LASER as the light source and measures the chemical concentration based on detection of a variation of LASER beam intensity after transmission along the optical path.

Laser absorption spectroscopy widely used for,

1. Intra-cavity LASER absorption spectroscopy
2. Cavity ring-down spectroscopy
3. Doppler-free Spectroscopy or saturation spectroscopy

16.1 Intra-cavity Laser Absorption Spectroscopy

Absorption measurements with particularly high sensitivity can be achieved with intra-cavity LASER absorption spectroscopy, where a sample is placed inside a LASER resonator. Typically, one records the LASER output spectrum at a certain time after turning the pump source on. The LASER would normally exhibit broadband emission with a smooth spectrum, but absorption lines create dips in the spectrum. Due to the large number of resonator roundtrips within the build-up time of radiation in the LASER, the method can be very sensitive.

16.2 Cavity ring-down spectroscopy (CRDS)

Spectroscopic technique that enables measurement of absolute optical extinction by samples that scatter and absorb light. It has been widely used to study gaseous samples which absorb light at specific wavelengths, and in turn to determine mole fractions down to the parts per trillion level. The technique is also known as cavity ring-down LASER absorption spectroscopy.

A typical CRDS setup consists of a LASER that is used to illuminate a high-finesse optical cavity, which in its simplest form consists of two highly reflective mirrors. When the LASER is in resonance with a cavity mode, intensity builds up in the cavity due to constructive interference. The LASER is then turned off in order to allow the measurement of the exponentially decaying light intensity leaking from the cavity. During this decay, light is reflected back and forth thousands of times between the mirrors giving an effective path length for the extinction on the order of a few kilometers.

16.3 Doppler-Free Spectroscopy

A fundamental challenge in the spectroscopy of gases is the thermal movement of the gas molecules, which causes Doppler broadening of the optical transitions – often going far beyond the natural linewidth. However, there are various techniques of Doppler-free LASER spectroscopy, where such effects are eliminated.

Saturation spectroscopy allows one to overcome the Doppler broadening. If gas atoms (or molecules) are exposed to a narrow-band LASER beam which is tuned to an absorption transition, only those atoms will be excited for which the Doppler shift makes them resonant with the LASER beam. For example, if the optical frequency of the pump LASER is on the lower end of the Doppler-broadened absorption line, essentially only those atoms will interact with it which are moving towards the pump light with an appropriate velocity. Similarly, a counter-propagating probe beam with the same optical frequency can interact only with atoms moving in the opposite direction. Therefore, the pump beam has virtually no influence on the absorption probed with the probe beam. This changes, however, when the frequency of both beams is tuned to the line center, so that they interact with the same atoms: the absorption for the probe beam is then somewhat saturated (reduced) if the pump beam is sufficiently intense. Therefore, one can detect a narrow dip in the middle of the absorption line as recorded with the appropriate beam. The width of that dip is determined by the natural linewidth, which can be much smaller than the Doppler-broadened linewidth.

Another possibility is Doppler-free Fourier transform spectroscopy involving two frequency combs with slightly different comb line spacings.

Applications of Laser Absorption Spectroscopy

Methods of LASER absorption spectroscopy are commonly used for quantitative measurements of concentrations gases or vapours. But not limited to this, LAS based techniques are also employed for detecting the composition of liquids, solids [or plasma some application areas are summarized as follows.

1- Environmental Monitoring

The temporal and spatial distribution of greenhouse gases, such as CO_2, CH_4, O_3, etc., is of great concern to those who study climate change. Moreover, Detection of some other atmospheric constituents (NH_3, NO) is necessary for environmental a1ssessment and protection. Concentrations of the trace gases in the atmosphere are measured e.g. with LASER radar (LiDAR) methods in the context of atmospheric environmental monitoring. Similarly, pollutants can be detected in water, and concentrations of medically active substances can be measured.

2- Industrial Process Measurement and Control

Continuous monitoring of some iconic gases, e.g. O_2, CH_4, C_2H_4, HCl, HF, etc., are usually required in the production process. LAS is one of the most promising techniques currently for the detection of industrial process gases, benefiting from its non-contact, high sensitivity, fast response and robustness.

3- In-situ Monitoring of Pollution Emissions

LAS based sensors have been proved to be robust when working in the harsh environment. Very typical applications also involve detection of exhaust emissions from coal-fired power plants, metal smelter, pharmaceutical factory, etc.

4- Biology and Medicine

For example, detection and analysis of volatile compounds in exhaled breath represents an attractive tool for monitoring the metabolic status of a patient and disease diagnosis, since it is non-invasive and fast. LAS has become a powerful tool to measure concentrations of various substances in human breath, which helps people retrieve vital information on medical conditions.

5- Combustion Diagnosis:

Line-of-sight LAS has been validated to quantitatively measure the path-averaged temperature, species concentrations, pressure and velocity in the combustion fields with sufficiently high temporal resolution. Spatially resolved 1D and 2D distributions of the parameters in the reacting flows are enabled by the combination of LAS and hard-field tomography.

6- Security Applications:

The detection objects often include explosives, drugs, chemical warfare agents and toxic gases. This is very meaningful for national for early-warning in national defense and counter-terrorism.